

Concept Question 7.1 Observations of molecular clouds

- a) You want to write a proposal to study a molecular cloud in which stars are forming. At what wavelength should you make your observations?
- b) Fig. 1 shows the dark molecular cloud B68 observed at three different wavelengths λ , specifically $0.44\ \mu\text{m}$, $1.25\ \mu\text{m}$, $2.16\ \mu\text{m}$, but with the same orientation and spatial extent (i.e. the images are all the same size). Why does it appear as a black blob? And why does the observed cloud's size decrease with increasing wavelength?

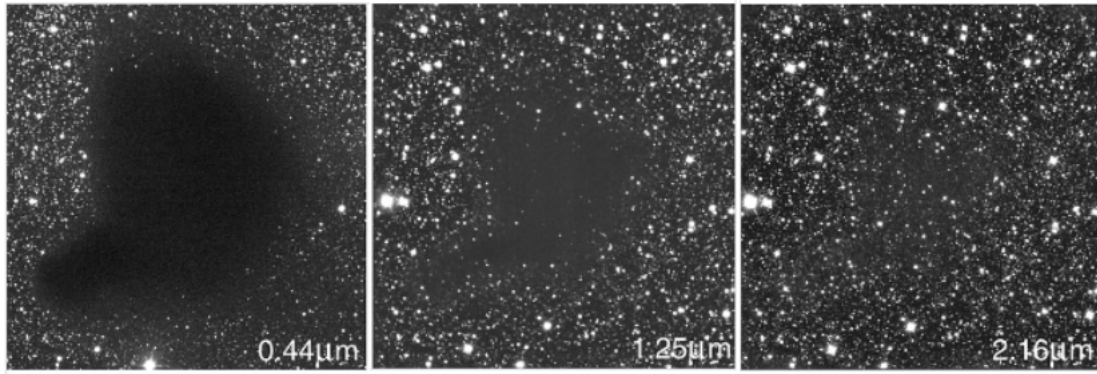


Figure 1: The dark cloud B68 at different wavelengths. Credit: ESO (adapted).

Problem 7.2 Timescale for contraction

In this exercise we will make a simple estimate of the timescale for the contraction of a molecular cloud. We will model the cloud as a spherically homogeneous cloud, and we will assume that the cloud suddenly loses its gas pressure support (e.g. due to cooling) and starts to contract under its own gravity.

- a) Use Newton's laws to derive the following expression for the infall velocity v as a function of the cloud radius r :

$$v(r) = -\sqrt{\frac{8\pi G r_0^2 \rho_0}{3} \left(\frac{r_0}{r} - 1 \right)}. \quad (1)$$

Hint: Use Newton's law of gravitation to express the gravitational force acting on a mass at radius r . Relate this force to the acceleration and the velocity at this radius using the chain rule. Integrate this relationship to find the above expression.

- b) Using Equation 1, derive an expression for the collapse timescale as a function of the initial density ρ_0 . What is the collapse timescale for a dense cloud of neutral hydrogen with $n_{\text{H}} = 10^6\ \text{cm}^{-3}$?

- c) In the previous points we have assumed that the cloud is homogeneous. How would you change this assumption to make it more realistic, and what would be the effect on the calculated timescale? And what could this imply for the star formation inside the cloud? Describe your answer qualitatively (no calculation required).

Problem 7.3 Numerical estimate of the Jeans mass

Consider a cold molecular cloud with temperature T and (particle) number density n . Assume the gas is predominantly molecular with mean molecular weight $\mu = 2.33$ (in units of m_{H}), and behaves as an ideal gas.

Given constants (cgs)

$$G = 6.67 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}, \quad k_{\text{B}} = 1.38 \times 10^{-16} \text{ erg K}^{-1}, \quad m_{\text{H}} = 1.67 \times 10^{-24} \text{ g}, \quad M_{\odot} = 1.99 \times 10^{33} \text{ g}.$$

The isothermal sound speed is

$$c_s = \sqrt{\frac{k_{\text{B}} T}{\mu m_{\text{H}}}}, \quad (2)$$

and the Jeans mass may be written as

$$M_J = \frac{\pi^{5/2}}{6} \frac{c_s^3}{G^{3/2} \rho^{1/2}}, \quad \rho = \mu m_{\text{H}} n. \quad (3)$$

- a) Compute c_s , ρ , and M_J for a typical molecular cloud:

$$T = 10 \text{ K}, \quad n = 10^3 \text{ cm}^{-3}.$$

Express M_J in solar masses.

- b) Repeat for a denser prestellar core:

$$T = 10 \text{ K}, \quad n = 10^5 \text{ cm}^{-3}.$$

- c) Using the scaling $M_J \propto T^{3/2} \rho^{-1/2}$, explain qualitatively why dense regions fragment more easily.
- d) Based on what you found in the previous point, what is more likely to form by gravitational cloud collapse? A Jupiter-mass or a Solar-mass object?

Problem 7.4 Virial theorem and molecular cloud stability

Consider a spherical molecular cloud of mass M , radius R , and one-dimensional velocity dispersion σ . Assume the cloud is isolated, non-rotating, and supported only by internal motions (thermal plus turbulent). Magnetic fields and external pressure are neglected.

a) Virial equilibrium

1. Starting from the scalar virial theorem,

$$2T + \Omega = 0, \quad (4)$$

write expressions for the kinetic energy T and gravitational potential energy Ω of a uniform-density cloud.

2. Show that the condition for virial equilibrium can be written as

$$\sigma^2 = \frac{1}{5} \frac{GM}{R}. \quad (5)$$

b) Virial parameter

The virial parameter is defined as

$$\alpha_{\text{vir}} \equiv \frac{5\sigma^2 R}{GM}. \quad (6)$$

1. Explain the physical meaning of $\alpha_{\text{vir}} < 1$, $\alpha_{\text{vir}} \simeq 1$, and $\alpha_{\text{vir}} > 2$.
2. Briefly discuss why observed giant molecular clouds (GMCs) in the Milky Way often have $\alpha_{\text{vir}} \sim 1$ –2.

c) Numerical plug-in: Milky Way GMC

Consider a Milky Way giant molecular cloud with:

$$R = 20\text{pc}, \quad \sigma = 3.0\text{km s}^{-1}.$$

Use $G = 4.302 \times 10^{-3} \text{ pc } M_{\odot}^{-1} (\text{km s}^{-1})^2$.

1. Compute the virial mass $M_{\text{vir}} = 5\sigma^2 R/G$.
2. Given an observed mass $M_{\text{CO}} = 1.5 \times 10^5 M_{\odot}$, compute the virial parameter and state whether the cloud is bound, marginally bound, or unbound.
3. Compute the surface density $\Sigma = M_{\text{CO}}/(\pi R^2)$ and compare to the typical GMC range $\Sigma \sim 50$ – $200 M_{\odot} \text{ pc}^{-2}$.
4. (Optional) Compute the turbulent crossing time $t_{\text{cross}} = R/\sigma$ and compare to the free-fall time. Comment briefly on cloud support versus collapse.